2024 AMC 10B Problem 14 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 14. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 14

Problem:

A dartboard is the region B in the coordinate plane consisting of points (x, y) such that $|x| + |y| \leq 8$. A target T is the region where $\left(x^2 + y^2 - 25\right)^2 \leq 49$. A dart is thrown and lands at a random point in B. The probability that the dart lands in T can be expressed as $\frac{m}{n} \cdot \pi$, where m and n are relatively prime positive integers. What is $m + n$?

Answer Choices:

A. 39

B. 71

C. 73

D. 75

E. 135

Solution:

The region B is a square with intercepts $(\pm 8, 0)$ and $(0, \pm 8)$. The area of this square is $(8\sqrt{2})^2 = 128$. Taking square roots shows that region T is the set of points that satisfy

$$25 - 7 \leq x^2 + y^2 \leq 25 + 7$$

which is an annulus (ring) with inner radius $\sqrt{18}$ and outer radius $\sqrt{32}$. Its area is $32\pi - 18\pi = 14\pi$. Note that T is internally tangent to B at the four points $(\pm 4, \pm 4)$. The required probability is the ratio of the area of T to the area of B, namely $\frac{14\pi}{128} = \frac{7}{64}\pi$. The requested sum is $7 + 64 = \text{(B)}\boxed{71}$.